

Security-Aware Workload-State Counterfactual Verification and Nodal Net-Value Settlement for Spatially Coupled AI Data Centers

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Abstract—Artificial-intelligence (AI) data centers are flexible grid resources, but a meter-only demand-response baseline cannot show that credited work was committed before the event, nor can it distinguish a local reduction from spatial rebound. This paper develops an auditable submit–execute–settle verifier that carries one declared workload state from counterfactual scheduling to signed nodal settlement. A sparse cumulative linear program and an exact job-indexed witness enforce release, deadline, energy, graphics-processing-unit (GPU) nameplate, and regional-capacity constraints. The reported risk profile is a validation-fitted simplex under total and daily conditional-value-at-risk budgets; a separate submission-masked gate policy and a fixed/flexible power-conversion certificate keep execution telemetry out of contract formation. On the 121-day trace inventory, the locked mechanism-isolation panel gives nRMSE 0.314, false credit 1.809 MWh/day, and $F_1 = 0.470$, compared with 0.339, 1.823 MWh/day, and 0.418 for the selected single projection. The ledger contains 71,144 submissions and 71,128 matched executions; its indexed witness has 12,296,675 job-slot variables and a declared-energy residual below 10^{-15} MWh. A 54-day gate replay uses only pre-event declarations, while an independent network replay evaluates the same witness under all 37 finite RTS-24 contingencies and two homogeneous scale transforms. Fixed facility demand is carried separately, so the raw q_{99} endpoint remains an auditable activation scenario rather than a silently clipped stress case. The certificate reports workload feasibility, information provenance, and signed N-1 value without treating public traces as co-located utility measurements.

Index Terms—AI data centers, counterfactual verification, demand response, security-constrained dispatch, nodal settlement.

I. INTRODUCTION

The growth of large-model training and inference is converting data centers from nearly static industrial loads into large, software-defined grid resources. Interactive inference is latency constrained, elastic inference can be delayed briefly, and batch graphics-processing-unit (GPU) work can be deferred for hours or migrated between regions. These capabilities enable demand response (DR), but they also invalidate the usual interpretation of a meter decrease. A reduction at one facility can be followed by delayed consumption or simultaneous rebound at another node. Paying the positive local decrease can therefore reward a redistribution that creates little or negative network value. Public traces already expose multi-week bursty workload arrivals and tens of thousands of scheduler jobs, while facility telemetry reports GPU energy at a different measurement boundary [1], [2].

Conventional DR measurement and verification estimates the unobserved no-event demand from historical meter data. Arithmetic historical baselines remain common in market practice under published market guidance [3], despite known bias and manipulation incentives [4]. Regression-based facility models quantify baseline uncertainty but do not encode a computing workload ledger [4]. More broadly, DR research establishes the economic role of responsive load, yet a meter-only estimator cannot determine whether queued jobs could actually have produced its predicted trajectory.

Data-center scheduling studies provide the missing physical state. Temporal deferral and geographical load balancing can reduce energy cost and peak demand [5], [6]. Recent work assesses data-driven flexibility from production traces and system-level security-constrained coordination, while balancing-market and clean-computing studies optimize spatial-temporal data-center dispatch [7], [8]. A recent power-system review also identifies quality-of-service (QoS), service-level-agreement (SLA), and market-participation constraints that are absent from meter-only accounting; clean-computing coordination provides a complementary dispatch context [9], [10]. Hierarchical optimal-consumption models further connect facility efficiency with response quality. Virtual-link market clearing and space-time remuneration establish how geographical and temporal transfers can enter electricity markets [11], [12]. Receding-horizon operation and uncertainty-aware frequency-regulation coordination further establish the need to retain future arrivals and multi-site state [13]. These methods optimize operator-controlled schedules; verification of an independently submitted DR counterfactual against an auditable workload ledger remains unresolved. This separation between computing feasibility and market verification motivates the auditable ledger used here. Recent production-trace studies quantify data-center flexibility and load-aggregator coordination, but leave this ledger-to-settlement coupling open [14], [15]. In particular, cross-regional dispatchable-capacity evaluation and coupled-regulation modeling quantify operational flexibility but do not provide a provenance-bound settlement certificate [16], [17]. Recent whole-facility measurements likewise require a separate bottom-up conversion from GPU power to facility demand; the present study therefore keeps raw energy calibration, benchmark scaling, and network placement as separate evidence layers.

The second gap is electrical. Geographic workload control is commonly valued with energy prices or gross megawatts. Such

accounting ignores the fact that nodal marginal value changes when congestion or a contingency constraint binds. Direct-current (DC) power flow and linear distribution factors provide transparent network sensitivities [18], [19]; N-1 dispatch can impose every credible post-contingency line limit through line-outage distribution factors (LODFs) [20]. No existing data-center DR verification framework jointly certifies workload feasibility, preserves signed spatial response, and settles the resulting security-aware network value. Alternating-current (AC) feasibility is a distinct nonlinear check, not a consequence of the DC certificate. Public workload and facility telemetry can support an observational replay of that ledger, but it does not constitute a causal utility event; a rigorous study must therefore separate measured-execution replay, counterfactual construction, and network-value verification to avoid conflating them with one field intervention.

This paper makes three power-system contributions. 1) It defines a typed submit–execute–settle evidence chain: an exact job-indexed counterfactual is formed only from scheduler declarations (release, runtime, requested GPUs, and declared energy entitlement), whereas the execution ledger is retained for post-event scoring and the settlement ledger is generated only after the coupling certificate passes. The indexed witness is independently rechecked before its regional profile is valued in the N-1 model. 2) It formulates a workload-state verifier with an explicit information boundary and a validation-fitted total-plus-CVaR risk fit, while keeping the pointwise false-credit cap in a separately auditable payment profile. A separately parameterized event policy and a binding-capacity stress cohort test whether the declaration-based feasible set remains nonempty when regional limits, service floors, and declared windows are jointly active. 3) It develops signed space–time nodal remuneration and a payment-activation rule over held-out power-conversion factors: the raw q_{99} endpoint is retained, but a payment can activate only when the precommitted fixed/flexible capacity certificate covers it. The central result is a dimension-preserving coupling invariant that carries one declared workload state into physical verification, network valuation, and settlement; the individual linear-program (LP), dispatch, and allocation primitives are standard and are not claimed as new in isolation. Because public traces do not identify utility geography, spatial conclusions are conditional on the declared feature- stratified trace scenario and the complete permutation/penetration panel. The certificate assumes trusted scheduler and telemetry provenance: a digest detects post-commitment record changes but cannot authenticate semantically false fields.

The remainder of the paper is organized as follows. Section II defines the spatial workload and network settlement problem. Section III presents the verifier, security-aware value rule, and allocation mechanism. Section IV specifies the data, baselines, and locked protocol. Section V reports the core counterfactual, execution-replay, job-level, security, payment, continuous-horizon, and provenance experiments. Section VI concludes with the deployment boundary and future validation requirements.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. Spatial workload state

Let $\mathcal{S}$, $\mathcal{K}$, $\mathcal{D}$, and $\mathcal{T}$ denote source regions, workload classes, destination data centers, and 15-minute intervals; the symbol h is reserved for a calendar day in statistical panels. The measured ledger supplies arrival energy a_{skt} for source s , class k , and release interval t . The decision $x_{skd\tau} \geq 0$ is the energy served at destination d in interval τ . Its cumulative state is

$$y_{skt} = \sum_{\tau=0}^t \sum_{d \in \mathcal{D}} x_{skd\tau}. \quad (1)$$

With cumulative arrivals $A_{skt} = \sum_{\tau=0}^t a_{sk\tau}$ and deadline D_k , release and deadline feasibility are

$$y_{skt} \leq A_{skt}, \quad (2)$$

$$y_{skt} \geq A_{sk,t-D_k}, \quad t \geq D_k. \quad (3)$$

The state recursion equals service, terminal service equals total arrivals, and $\sum_{s,k} x_{skdt} \leq \bar{P}_d \Delta t$. Facility demand is

$$p_{dt} = P_d^{\text{fix}} + \Delta t^{-1} \sum_{s,k} x_{skdt}. \quad (4)$$

For rolling window end T^+ and final event-day release T^0 , the terminal state is not forced to consume unexpired future work. Instead,

$$y_{skT^+} = A_{sk, \max\{T^0, T^+ - D_k\}}, \quad (5)$$

which completes every job due inside the window and every event-day arrival, while retaining later real arrivals in the release constraints. The window starts $D_{\max} + 96$ intervals before the event day and ends $D_{\max}$ intervals after it; omitted earlier work must therefore expire before the evaluated day. Equations (2) and (3) prevent early execution and late completion, while (4) maps the feasible service state to the facility meter. The variable x is an aggregate, preemptive service envelope: it is valid for checkpointable batch work and does not by itself assert a nonpreemptive placement for every GPU job. For the measured MIT replay, each joined job is also retained with its contiguous execution interval, runtime, GPU count, and native average power; Experiment 14 checks that witness independently. This two-level construction keeps counterfactual optimization sparse while making the task-level scope explicit. For the counterfactual task-level check, let $j \in \mathcal{J}_{\text{sub}}$ index a submitted scheduler record, r_j denote its submitted release, and let d_j be the service-window endpoint. Slurm “timelimit” is an allocation run-time declaration; it is converted to $R_j = \lceil \text{timelimit}_j / \Delta t \rceil$ slots and combined with a precommitted queue allowance $B = 96$ slots, so $d_j = r_j + R_j + B$. The unlimited Slurm sentinel is mapped to the precommitted horizon $H_{\infty} = 128$ slots before adding B ; finite declarations are retained exactly. The 0.001-MW-per-GPU nameplate is an audit-only feasibility precheck: if a declared window cannot serve $\hat{E}_j$, the run fails closed and the window is not enlarged. Thus d_j is available at submission and never uses the observed completion time. Let $\hat{E}_j$ be the training-calibrated declared service entitlement, s_j its submit-time scenario region, and g_j^{req} its requested GPU

count. The indexed service variable $u_{j\tau}$ is constrained by

$$\begin{aligned} u_{j\tau} &\geq 0, & \sum_{\tau=r_j}^{d_j-1} u_{j\tau} &= \hat{E}_j, \\ u_{j\tau} &\leq g_j^{\text{req}} \bar{p}_{\text{GPU}} \Delta t, \\ \sum_{j:s_j=d, r_j \leq \tau < d_j} u_{j\tau} &\leq \bar{P}_d \Delta t, \\ \min_u \sum_{j,\tau} [w_{\text{batch}}(\tau - r_j) + r^{\text{DR}} \mathbf{1}\{\tau \in \mathcal{E}\}] u_{j\tau}. \end{aligned} \quad (6)$$

Here $\bar{p}_{\text{GPU}}$ is the precommitted 0.001 MW/GPU nameplate. Thus the counterfactual is a globally solved, job-indexed preemptive batch schedule with explicit release, submit-time service window, energy, site capacity, and GPU-count constraints. It is not obtained by replaying the measured profile; its event tariff is fixed before scoring. Experiment 19 solves (6) for all 71,144 submitted jobs; 71,128 have a matching positive-energy execution record for the independent score. The observed $[t_j^{\text{start}}, t_j^{\text{end}}]$ interval is retained only to build the independent native replay used for scoring and never constrains $u_{j\tau}$. For the job-to-network certificate, the indexed witness is mapped without a second optimization. Let $\mathcal{J}_{skd}$ be the ledger partition whose source, class, and declared destination labels are s, k, d ; Experiment 19 uses the native assignment $d = s_j$ and therefore has no outcome-dependent migration:

$$\begin{aligned} x_{skd\tau}^{\text{job}} &= \sum_{j \in \mathcal{J}_{skd}: r_j \leq \tau < d_j} u_{j\tau}, \\ p_{d\tau}^{\text{job}} &= P_d^{\text{fix}} + \Delta t^{-1} X_{d\tau}^{\text{job}}, \quad X_{d\tau}^{\text{job}} = \sum_{s,k} x_{skd\tau}^{\text{job}}. \end{aligned} \quad (7)$$

Aggregate migration remains a matched-workload control; (8) is the indexed replay profile. Experiment 22 reconstructs X^{job} from every stored $u_{j\tau}$, rechecks the indexed primal, and verifies equality before network valuation, so no separately optimized aggregate trajectory receives a network value. Workload deferral and geographic assignment follow [7], [8], while rolling-state control follows [13]; waiting and migration costs are $w_k(\tau - t)x_{skd\tau}$ and $m_{sd}x_{skd\tau}$, with service-time substitution differing only by a fixed-arrival constant.

For provenance, the submit-time ledger and post-event execution ledger have separate canonical digests. Each submitted record is represented by the scheduler-only tuple and its digest in

$$\begin{aligned} \ell_j^{\text{sub}} &= (\text{id}_j, t_j^{\text{submit}}, \text{timelimit}_j, q_j, \kappa_j, \sigma_j), \\ \mathcal{L}_{\text{sub}} &= \text{sort}\{\ell_j^{\text{sub}} : j \in \mathcal{J}_{\text{sub}}\}, \\ h_{\text{sub}} &= \text{SHA256}(\text{serialize}(\mathcal{L}_{\text{sub}})). \end{aligned} \quad (9)$$

where q_j , κ_j , and σ_j denote the requested-resource, class, and scheduler-state fields. The execution ledger separately stores start/end times, DCGM energy, and telemetry counts; its one-to-one join, release-execution order, and raw-to-join energy equality are checked only for post-event reconciliation. The Secure Hash Standard [21] and canonical ordering make h_{sub} in (9) independent of execution telemetry. Experiment 16 reconciles the measured envelope with the 118-MW nameplate fixed before the validation/test split.

The verifier treats scheduler/DCGM services as authoritative and assumes the digest is committed before the event decision.

It cannot establish that a deadline, region, class, or energy field was truthful before signing; that requires upstream attestation and lies outside this optimization certificate.

B. Counterfactual, response, and false credit

For event set $\mathcal{E} \subset \mathcal{T}$, p^0 is the no-event counterfactual, $p^{1,\text{sim}}$ is the workload-optimized event trajectory, and p^{obs} is the independently measured MIT/DCGM execution trace. Because no utility event label is present, p^{obs} is an observational alignment target and never a causal outcome or pre-decision input. In mechanism-isolation rows $r = p^0 - p^{1,\text{sim}}$ is signed, so rebound is retained. For source $z \in \{\text{obs}, \text{sim}\}$, submitted and true credit are $\hat{r}^{(z)} = [\hat{p}^0 - p^{(z)}]_+$ and $r^{(z)} = [p^0 - p^{(z)}]_+$; source-specific false credit is

$$F^{(z)} = \Delta t \sum_{d,t \in \mathcal{E}} [\hat{r}_{dt}^{(z)} - r_{dt}^{(z)}]_+, \quad z \in \{\text{obs}, \text{sim}\}. \quad (10)$$

The source-specific quantity in (10) is evaluated separately for the simulated and observed trajectories. The simulated source defines the validation epigraph; the observed source is recomputed only in the locked replay. For nonnegative true credit, $p^{(\text{sim})} + r^{(\text{sim})} = \max\{p^0, p^{(\text{sim})}\}$, an audited identity rather than a new meter trajectory. The gross forecast credit is not a transfer: before the event, the operator freezes p^{con} , and after meter closure the payable response is the pointwise intersection

$$Q^{\text{pay}} = \Delta t \sum_{d,t \in \mathcal{E}} \min\{[\hat{p}_{dt}^0 - p_{dt}^{\text{obs}}]_+, [p_{dt}^{\text{con}} - p_{dt}^{\text{obs}}]_+\}. \quad (11)$$

Equation (11) uses only the submitted baseline, frozen contract, and closed meter. The trace-anchored p^0 is an offline over/underpayment diagnostic and never enters fitting, gate decisions, workload optimization, or transfer formation. The study reports gross, payable, oracle over/underpayment, nRMSE, bias, precision, recall, and F_1 .

C. Strategic reference scheduling

Consider an arithmetic mean of $H = 10$ reference days, event probability q , and DR price π^{DR} . If reference day j adds event-window service δR_j , the expected payment change is

$$\Delta \Pi = \frac{q\pi^{\text{DR}}}{H} \sum_{j=1}^H \delta R_j - \sum_{j=1}^H \Delta C_j. \quad (12)$$

Let c'_j be the right derivative of the minimum operating cost with respect to an event-window minimum-service constraint on day j . Linear-program sensitivity [22] gives the following exact local condition:

$$\frac{q\pi^{\text{DR}}}{H} > \min_j c'_j \iff \exists \text{ a strictly profitable local deviation.} \quad (13)$$

At a degenerate optimum the dual at the current service level can be zero even when the right-direction marginal cost is positive. Experiment 1 therefore certifies c'_j with two independent right finite differences of 5×10^{-4} and 10^{-3} MWh and a 10^{-6} USD/MWh agreement tolerance. The historical-baseline premise is documented in [4]; (12)–(13) are this paper's workload-specific specialization.

D. Network value

Let M be the data-center-to-bus incidence matrix and $p_t = p_t^{\text{native}} + Mp_t^{\text{dc}}$ the nodal demand vector. For generator-to-bus map C_g , generator bounds, and power-transfer distribution factor (PTDF) H with line ratings $\bar{f}$, the piecewise-linear security-constrained economic dispatch (SCED) value is

$$\begin{aligned} V(p) = \min_g \sum_i C_i(g_i) \\ \text{s.t. } \mathbf{1}^\top g = \mathbf{1}^\top p, \quad \underline{g} \leq g \leq \bar{g}, \\ -\bar{f} \leq H(C_g g - p) \leq \bar{f}. \end{aligned} \quad (14)$$

The incidence mapping aggregates device demand at its connection bus, and (14) is the standard lossless DC representation [18], [19]. The realized signed grid value is

$$\Delta V = V(p^0) - V(p^1). \quad (15)$$

Positive ΔV is a credit and negative ΔV a debit; no positive-only truncation is applied.

III. PROPOSED METHODOLOGY

A. Model-based framework and matched-information pre-estimation

The state, power, and network equations in Section II are the first stage of the methodology: they define the feasible workload state before any statistical target or payment is formed. Fig. 1 shows the complete chain. Meter history and the submitted-job ledger enter a statistical pre-estimator and the workload feasible set in parallel. The pre-estimator never receives execution energy used as scoring truth. For each event day, it uses only the information available under one of two declared protocols: lagged meter/calendar features for online baselines, or the complete submitted ledger for equally informed post-event comparisons. Ridge regression and gradient boosting are used as standard matched-information comparators [23], [24]; extremely randomized trees and synthetic control provide two additional nonlinear and donor-pool comparators [25], [26]. Two stronger comparators are also solved: median-loss gradient boosting with the complete submitted ledger and a nonnegative synthetic control whose weights globally minimize pre-event absolute error over 28 donor days. To separate the effect of projection from that of pre-estimation, the complete-ledger quantile output is also passed through the same exact workload-feasible projection, with its penalty chosen by contiguous validation folds. The same scheduler declaration view is supplied to the post-event learner, the selected single projection, and the proposed convex verifier; execution energy and completion timestamps are withheld until the locked scoring pass.

B. Sparse cumulative workload-state verifier

For statistical estimate $\tilde{p}$ and each predeclared penalty ρ_ℓ , the first-stage program solves

$$\min_{x,y,z} C_w(x) + \rho_\ell \sum_{d,t} z_{dt} \quad \text{s.t.} \quad z_{dt} \geq \pm(p_{dt}(x) - \tilde{p}_{dt}), \quad (16)$$

subject to (1)–(4). Absolute-value epigraphs are standard convex constructions [22]; using them to project a meter baseline onto an auditable workload ledger is specific to this work. Here

C_w denotes workload operating cost; C_i denotes generator-segment cost in (14), and $V(p)$ denotes the network dispatch value. Thus (16) changes the objective, not the physical feasible set. Six globally solved schedules $p^{(\ell)}$ share identical arrivals and constraints. Their validation-only convex combination $\bar{p} = \sum_\ell \alpha_\ell p^{(\ell)}$, $\alpha \geq 0$, $\mathbf{1}^\top \alpha = 1$, remains feasible because the common scheduling set is convex.

The coefficients minimize validation squared error subject to both total and daily-tail mechanism-isolation false-credit budgets, where conditional value at risk (CVaR) measures the latter [27]. Let ℓ^* be the single candidate with minimum worst contiguous-fold nRMSE and define $B_j^{\text{cap}} = F_j^{(\text{sim})}(\ell^*)$, where $F_j^{(\text{sim})}$ is obtained from the explicit submitted-minus-true credit definition above. The observed-meter source is audited separately after the locked decision. For reserve $\beta \in (0, 1]$,

$$\sum_j F_j^{(\text{sim})}(\alpha) \leq \beta \sum_j B_j^{\text{cap}}, \quad (17)$$

$$\text{CVaR}_{0.75}(F_j^{(\text{sim})}(\alpha)) \leq \beta \text{CVaR}_{0.75}(B_j^{\text{cap}}). \quad (18)$$

For implementation, the positive part is not differentiated through a sort. For each event sample n and validation day j , the program introduces $f_n \geq 0$, a CVaR threshold $\nu \geq 0$, and tail slacks $u_j \geq 0$:

$$f_n \geq p_n^\alpha - p_n^{(\text{sim})} - r_n^{(\text{sim})}, \quad (19)$$

$$F_j^{(\text{sim})} = \sum_{n \in j} f_n, \quad (20)$$

$$F_j - \nu - u_j \leq 0, \quad (21)$$

$$\nu + \frac{1}{0.25J} \sum_j u_j \leq \beta \text{CVaR}_{0.75}(B_j^{\text{cap}}). \quad (22)$$

Here $r_n^{(\text{sim})} = [p_n^0 - p_n^{(\text{sim})}]_+$ is the true mechanism-isolation credit. The equivalent threshold identity is checked in the implementation, but the two credit terms remain explicit in the model. This epigraph is never substituted for the frozen contract profile p^{con} in (11). The explicit epigraph in (22) is the implementation used in all reported fits. Together with the simplex equality, box bounds on α , and the total budget in (17), these are linear constraints and the squared error objective is a convex quadratic program. HiGHS first supplies a feasible epigraph point; the full epigraph is then solved with a sparse primal–dual trust-region method. The returned stationarity residual, every linear epigraph residual, the simplex equality, and all bound residuals are recomputed from the final primal point. The optimizer status and these residuals are stored for every fit. Four contiguous folds select β using validation nRMSE, then risk ratio; held-out noninferiority is reported as an outcome and is not used as a hidden constraint. A separately reported payment-contract projection targets $\bar{p}$ while imposing, for every event sample,

$$p_{dt}^{\text{safe}} \leq p_{dt}^{(\ell^*)}. \quad (23)$$

The pointwise cap in (23) is the upper side of the contractual credit band; it is evaluated on every event slot before settlement and is not substituted for the scientific risk profile.

a) *Two-sided credit band.*: The upper envelope alone would admit an arbitrarily conservative zero-credit profile. The method therefore imposes a lower band tied to the

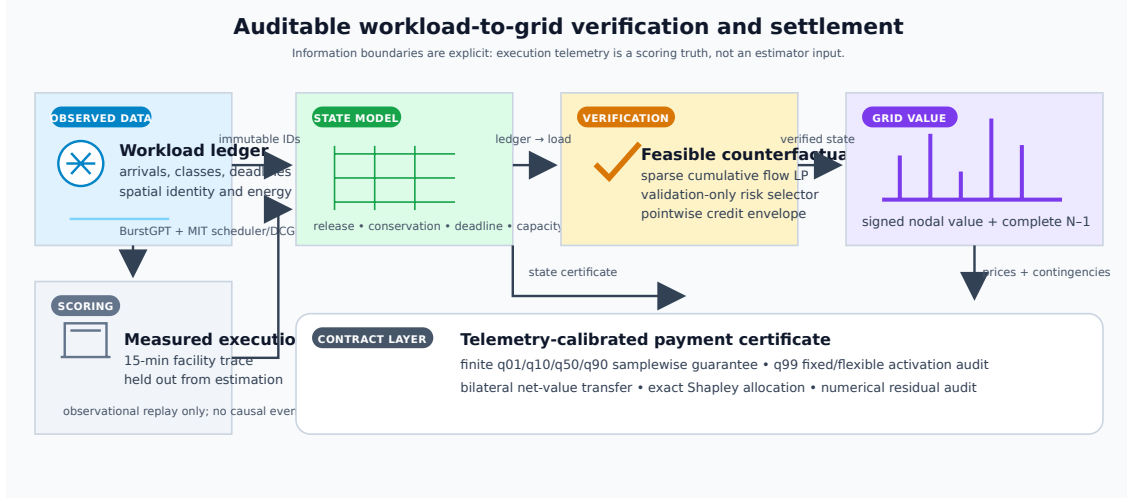


Fig. 1. Verification-to-settlement framework. Validation labels select the counterfactual risk contract; locked execution truth is used only for scoring. The right branch computes aggregate N-1 value and exact participant shares.

independently fitted convex target. Define the componentwise risk floor $p_{dt}^{\text{floor}} = \min\{p_{dt}^{(\ell^*)}, \bar{p}_{dt}\}$, where $\bar{p}$ is the validation-fitted convex target before the final workload projection. The event-slot constraints are

$$p_{dt}^{\text{floor}} - \varepsilon \leq p_{dt}^{\text{safe}} \leq p_{dt}^{(\ell^*)}, \quad (d, t) \in \mathcal{D} \times \mathcal{E}, \quad (24)$$

where $\varepsilon = 1$ MW is fixed before the locked test split. The lower bound is implemented together with the workload constraints, not as a post-processing clip. Thus the output remains an attainable schedule and can move below the single reference when the independent risk fit supports a safer trajectory. The total constraint (17) controls aggregate exposure and (18) controls the worst daily tail. The upper side is the deterministic payment cap; the lower side is a declared under-credit regularizer relative to p^{floor} , not a distribution-free guarantee against an arbitrary unseen meter trace. Overpayment against the realized event is handled by the contract-capped settlement rule (11). The selected single feasible projection proves nonemptiness, and the locked results report both margins and the post-event payable amount for this separate payment profile.

b) Decision-time information boundary.: The ex-post selector may use the complete ledger, but a gate deployment removes all post-gate arrivals before target construction and optimization. The target is refit with the matched-information metadata model on this masked ledger. A no-tariff LP freezes the contract baseline and commits 5 MWh of event service; a second LP applies the positive DR price to the same ledger and carries unexpired batch state to the next checkpoint. Reserved future jobs cannot be credited. Baseline error is computed only for the submitted contract; the tariff-bearing plan is scored as planned reduction against the closed meter and then settled by (11), keeping the two roles distinct.

To separate capacity planning from payment eligibility, the gate publishes a causal reserve envelope. For candidate quantile η and event day d ,

$$\hat{a}_{sk\tau}^{\text{res}}(d; \eta) = Q_{\eta}(\{a_{sk\tau}(h) : h < d\}), \quad \tau \geq t_{\text{gate}}, \quad (25)$$

Equation (25) uses temporally earlier days only. Candidates

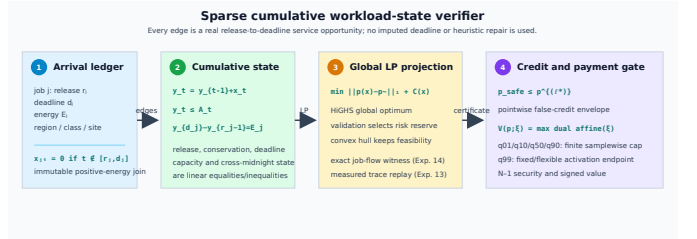


Fig. 2. Sparse cumulative verifier detail. Each admissible edge is a declared release-to-deadline service opportunity; the global projection, risk fit, and payment gate operate on the same feasible state and do not repair missing deadlines or execution records.

are selected on validation days under a declared false-credit budget and used only for capacity planning; no reserved job enters payment before ledger commitment.

The cumulative state makes prefix storage linear rather than quadratic in the horizon: explicit prefixes scale as $O(|\mathcal{S}||\mathcal{K}||\mathcal{D}||\mathcal{T}|^2)$, whereas the cumulative matrix scales as $O(|\mathcal{S}||\mathcal{K}||\mathcal{D}||\mathcal{T}|)$. A 512-slot deadline is handled by (5) with every observed future arrival retained; no zero-arrival completion buffer is used in the final continuous-horizon certificate. The day trajectory is embedded by a lexicographic pair of linear programs: minimum L1 distance first, followed by minimum operating cost on that optimal-distance face. Hence no penalty weight trades physical fit against cost. All linear programs are solved to global optimality with HiGHS [28]. The resulting state-to-credit sequence is summarized by the framework in Fig. 1; the execution replay and interval audit are reported separately in Experiments 13–15. Fig. 2 resolves the ledger-to-LP path at the component level.

C. Signed nodal and N-1 settlement

Let $\lambda^0 \in \partial V(p^0)$ be the nodal demand-value subgradient. Signed linear settlement is $(\lambda^0)^T(p^0 - p^1)$, while (15) is the realized system-value benchmark and the payoff of an explicit bilateral avoided-cost contract; it is not relabeled as

an independent system operator (ISO) market payment. The convex subgradient inequality [22] yields the global certificate $0 \leq (\lambda^0)^\top (p^0 - p^1) - \Delta V = V(p^1) - V(p^0) - (\lambda^0)^\top (p^1 - p^0)$. (26)

Thus the linear rule globally upper-bounds exact value in the implemented polyhedral market model. Equality holds only when both endpoints share a supporting dual subgradient.

For market-compatible remuneration, let $\mathcal{W}$ contain the complete response cycle from the earliest deadline-feasible anticipatory shift through the latest recovery interval. The space-time rule is

$$\Pi^{\text{ST}} = \Delta t \sum_{t \in \mathcal{W}} \sum_d \lambda_{dt}^0 (p_{dt}^0 - p_{dt}^1). \quad (27)$$

This is the nodal virtual-link remuneration structure of [11], [12], applied here to a verified counterfactual. Summing site payments reproduces (27) exactly.

Space-time energy remuneration is only one component of the contracted DR product. Let $\mathcal{D}^{\text{part}}$ be the predeclared participating sites, $\mathcal{E}$ the event intervals, and r^{DR} the cleared service price. The signed delivered capacity service and total operator value are

$$Q^{\text{cap}} = \Delta t \sum_{d \in \mathcal{D}^{\text{part}}} \sum_{t \in \mathcal{E}} (p_{dt}^0 - p_{dt}^1), \quad (28)$$

$$G = r^{\text{DR}} Q^{\text{cap}} + \Pi^{\text{ST}}, \quad (29)$$

where p^0 is the minimum-cost no-event schedule under the identical continuous arrival window. Capacity credits and ex-post participation constraints follow established DR mechanism design [29], [30]; the signed Π^{ST} term separately debits spatial transfer and recovery. Equations (28)–(29) jointly define the signed service and operator value.

Let x^1 and x^0 denote the workload service schedules inducing p^1 and p^0 , respectively. Define $C_{\text{opp}} = C_w(x^1) - C_w(x^0)$ as the participant opportunity cost and $S = G - C_{\text{opp}}$ as the transferable transaction surplus. With disagreement utilities equal to zero, the symmetric Nash solution [31] activates the contract iff $S \geq 0$ and sets

$$z = \mathbb{I}\{S \geq 0\}, \quad P^{\text{B}} = z \left(C_{\text{opp}} + \frac{S}{2} \right), \quad u_{\text{dc}} = u_{\text{op}} = z \frac{S}{2}. \quad (30)$$

Thus (30) makes voluntary nonactivation the explicit outside option. For every activated contract, both utilities are nonnegative and the single transfer P^{B} is exactly budget balanced. The bilateral transfer is reported separately from both ISO nodal remuneration and the exact dispatch value benchmark. Unlike event-only accounting, $\mathcal{W}$ debits both pre-event load increases and delayed rebound; workload conservation gives $\sum_{d,t \in \mathcal{W}} (p_{dt}^0 - p_{dt}^1) \Delta t = 0$ up to solver tolerance.

Security-aware settlement augments (14) with every finite non-islanding single-line outage. If $L_{\ell k}$ is the LODF from outage k to monitored line ℓ , then

$$-\bar{f}_\ell \leq (H_\ell + L_{\ell k} H_k)(C_{g\ell} g - p) \leq \bar{f}_\ell, \quad \ell \neq k. \quad (31)$$

Equation (31) is the standard preventive N-1 LODF formulation [20]; the experiment imposes all credible rows simultaneously, with no contingency screened out. Native public ratings are retained. Islanding outages are reported and excluded because a connected single-reference PTDF cannot represent separate

islands, consistent with the scope of DC security models and the N-1 planning criterion [32].

D. Scenario-robust N-1 payment certificate

The pointwise envelope controls credited energy but does not by itself certify network payment because spatially different demand vectors can activate different security constraints. Let $\mathcal{P} = \text{conv}\{p^{(1)}, \dots, p^{(L)}\}$ be the convex hull of complete workload-feasible schedules and let $p^{\text{cap}} \in \mathcal{P}$ be the validation-selected single feasible projection. Its penalty ρ_ℓ is selected by contiguous validation and is reported separately from the payment-target penalty. The feasible-quantile projection remains an external transfer comparator. Let Ξ_5 contain the 1st, 10th, 50th, 90th, and 99th percentiles of held-out measured-to-predicted GPU energy; these predeclared scenarios are supported by independent telemetry [2]. For facility profile p_{dt}^{fac} , fixed demand P_d^{fix} , and benchmark scale s_{dc} , conversion factor ξ gives

$$p_{dt}^{\text{dc},\xi} = s_{\text{dc}} (P_d^{\text{fix}} + \xi [p_{dt}^{\text{fac}} - P_d^{\text{fix}}]), \quad p_{dt}^{\text{fac}} \geq P_d^{\text{fix}}. \quad (32)$$

The scale is calibrated on the held-out q_{99} flexible peak, so the fixed component is invariant to ξ and both raw endpoints remain un-clipped activation scenarios; the flexible-only factor in Experiment 16 is diagnostic. In what follows, $Mp_t^{\text{dc},\xi}$ is the nodal injection and superscripts α and “cap” apply (32) to p^α and p^{cap} , respectively. For each settlement day, the payment-certified verifier defines $B_\xi^{\text{cap}} = \Delta t \sum_{t \in \mathcal{E}} V_{N-1}(p_t^{\text{native}} + Mp_t^{\text{dc},\xi,\text{cap}})$ and solves

$$\begin{aligned} & \text{lexmin}_{\alpha, e, \eta, \{g_t^\xi\}} \left(\sum_{d,t \in \mathcal{E}} e_{dt}, -\eta \right) \\ & \text{s.t.} \quad p^\alpha = \sum_{\ell=1}^L \alpha_\ell p^{(\ell)}, \\ & \quad \alpha \geq 0, \quad \mathbf{1}^\top \alpha = 1, \quad 0 \leq \eta \leq 1, \\ & \quad e_{dt} \geq \pm (p_{dt}^\alpha - p_{dt}^{\text{safe}}), \\ & \quad g_t^\xi \text{ satisfies (14) and (31),} \\ & \quad \text{with } p_t = p_t^{\text{native}} + Mp_t^{\text{dc},\xi,\alpha}, \\ & \quad \quad \quad t \in \mathcal{E}, \quad \xi \in \Xi_5, \\ & \quad \Delta t \sum_{t \in \mathcal{E}} \sum_i C_i(g_{it}^\xi) \\ & \quad \leq (1 - \eta) B_\xi^{\text{cap}}, \quad \xi \in \Xi_5. \end{aligned} \quad (33)$$

Generator costs use the same declared piecewise-linear epigraph as market settlement. Two global linear programs implement the lexicographic objective: the first obtains the minimum trajectory error, and the second preserves that optimum within solver tolerance while maximizing the common fractional margin η . Existence of the primal dispatches in (33) is equivalent to the corresponding upper bound on the optimal N-1 value; the unit weight on p^{cap} proves nonemptiness in every scenario. Consequently, for any $\xi \in \Xi_5$ and realized event load $p^{1,\xi}$, daily payment obeys

$$\begin{aligned} & \sum_{t \in \mathcal{E}} \Delta t [V_{N-1}(p_t^{\text{native}} + Mp_t^{\text{dc},\xi,\alpha}) - V_{N-1}(p_t^{1,\xi})] \\ & \leq \sum_{t \in \mathcal{E}} \Delta t [V_{N-1}(p_t^{\text{native}} + Mp_t^{\text{dc},\xi,\text{cap}}) - V_{N-1}(p_t^{1,\xi})]. \end{aligned} \quad (34)$$

The common realized-cost term cancels, so the certificate requires no execution truth and holds samplewise. Both programs contain every finite non-islanding contingency. The formulation follows the standard SCED primal and linear-program epigraph foundations [19], [22]. Here $\mathcal{P}$ contains the six independently solved first-stage projections from (16) and the matched feasible-quantile projection. The reported risk-constrained verifier is the validation-fitted simplex point in this hull under the total and CVaR budgets. The separately stored payment-contract profile is projected toward that risk target under the pointwise activation cap; it is not fed back into the risk fit. The quantile schedule remains an independent transfer comparator.

The finite cap is complemented by a set-valued payment certificate. The contractual cap profile is denoted $p^{\text{cap}} = p^{(\ell^*)}$, where ℓ^* is selected by contiguous validation nRMSE; the independent payment-target profile p^{pay} is selected separately by validation payment error and is used only as the target of the certificate. For a realized counterfactual value, define $\Delta V_\xi(p) = V_{N-1}(p, \xi) - V_{N-1}(p^1, \xi)$ and the continuous segment $p_\theta = (1 - \theta)p^{\text{cap}} + \theta p^{\text{cert}}$, $0 \leq \theta \leq 1$. The induced interval is

$$\mathcal{I}_\xi = \left[\min_{0 \leq \theta \leq 1} \Delta V_\xi(p_\theta), \max\{\Delta V_\xi(p^{\text{cap}}), \Delta V_\xi(p^{\text{cert}})\} \right]. \quad (35)$$

The lower endpoint is obtained by one exact joint N-1 SCED LP with a shared segment parameter; the upper endpoint follows from convexity of the optimal dispatch value. The selected cap payment is an element of $\mathcal{I}_\xi$. The endpoints use only validation-frozen feasible profiles and the realized counterfactual needed for settlement; no oracle baseline selects or certifies the interval. Interval width is reported as uncertainty cost, while the seven-profile hull remains the finite-scenario payment certificate in (33).

E. Participant allocation

For participants N , define $v(S)$ as signed dispatch value when coalition S moves from baseline to actual demand and nonmembers remain at baseline. The exact allocation is

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [v(S \cup \{i\}) - v(S)]. \quad (36)$$

This is the Shapley value [33]; the characteristic function based on signed grid value is the proposed specialization. Grouping (36) by coalition size cancels every intermediate value and gives $\sum_i \phi_i = v(N)$ when $v(\emptyset) = 0$. All 16 coalitions for four sites and all 256 coalitions for eight contractual portfolios are enumerated; no permutation estimator is used. For larger auditable contracts, exchangeable slices at the same electrical site are grouped by their member count. The exact sum retains the binomial multiplicity of every labeled coalition and evaluates only $\prod_g (m_g + 1)$ distinct count states. This symmetry reduction remains the Shapley value and is evaluated through 20 participants.

F. Analytical guarantees

The following certificates make the logical scope explicit. First, under the interior and differentiability assumptions of

(12), a feasible reference-day perturbation is strictly profitable if and only if $q\pi^{\text{DR}}/H > \min_j c'_j$ in (13); Experiment 1 obtains the right-hand-side duals and checks the independently solved 56-cell strategic grid. The workload set is linear and therefore convex. Every simplex schedule and the two-sided envelope in (24) are feasible, and for any execution c ,

$$\begin{aligned} [p_{dt}^{\text{safe}} - c_{dt}]_+ &\leq [p_{dt}^{(\ell^*)} - c_{dt}]_+, \\ [c_{dt} - p_{dt}^{\text{safe}}]_+ &\leq [c_{dt} - p_{dt}^{\text{floor}}]_+ + \varepsilon. \end{aligned}$$

Summation gives the false-credit and under-credit bounds without using locked outcomes.

Second, the base-case and stacked N-1 dispatch values are convex and piecewise affine in nodal demand. The subgradient inequality is exactly (26), with $0 \leq \Delta V_{\text{lin}} - \Delta V \leq \|\lambda^1 - \lambda^0\|_* \|p^1 - p^0\|$; positive-only settlement has no such identity. In (33), each embedded dispatch is a feasible witness, so subtracting the common realized cost gives (34) for every $\xi \in \Xi_5$. The dual representation also makes $V_{N-1}(p; \xi)$ convex in a continuous conversion factor, so the worst case on $[\xi_L, \xi_U]$ is attained at an endpoint. The lower endpoint of (35) is the exact joint-LP minimum over the declared affine workload segment and convexity makes its endpoint maximum exact; Experiments 9 and 15 use independent 40- and four-segment models.

Finally, collision resistance of SHA-256 makes post-commit changes to the scheduler-only tuple detectable in h_{sub} , while execution joins check ordering and energy conservation only after the event. An optimum of (6) gives each submitted job its entitlement within its submit-time window, enforces site capacity and the requested-GPU nameplate, and does not rely on aggregate rounding; contiguous execution is audited separately. With $B_{dj} = \mathbf{1}\{s_j = d\}$ and bus map M , the job, aggregation, and mapping residuals are

$$\begin{aligned} r_j^{\text{job}} &= \sum_{\tau=r_j}^{d_j-1} u_{j\tau} - \hat{E}_j, \\ r_{d\tau}^{\text{agg}} &= X_{d\tau} - \sum_j B_{dj} u_{j\tau}, \\ r_{b\tau}^{\text{map}} &= p_{b\tau} - P_{b\tau}^{\text{fix}} - \Delta t^{-1} \sum_d M_{bd} X_{d\tau}. \end{aligned}$$

If their typed norms are at most ε , the dispatched profile is the same indexed witness up to that tolerance; at zero residual an independently optimized aggregate cannot receive settlement value. Experiment 22 checks these residuals before N-1 valuation and preserves $\sum_i \phi_i = v(N)$.

IV. EXPERIMENTAL DESIGN

A. Data, preprocessing, and information boundary

The complete data inventory is as follows. BurstGPT supplies 5,188,507 successful requests and token service [1]; MIT Supercloud supplies scheduler submissions and independent Data Center GPU Manager (DCGM) energy. Deterministic validation leaves 71,144 submitted jobs and 71,128 positive-energy execution matches, of which 68,664 start in the common 121-day tensor used by Experiments 1-13 [2]. Submission time defines arrivals and measured intervals define scoring truth. A held-out GPU-power model (28,413 observations) gives

$R^2 = 0.899$, MAE 11.22 W, and RMSE 15.82 W; workload scaling is fitted only on the first 40 complete training days. The held-out per-job energy ratios define Ξ_5 in (33) without synthetic replacement. Experiment 14 extends the replay to the last completion and retains all 71,128 matches, whereas Experiment 19 uses only scheduler declarations and a training-only entitlement; the execution join is opened only for post-event scoring. Its runtime window plus the precommitted 96-slot queue allowance is a contract-window definition, not a submit-to-completion interpretation of Slurm “timelimit”.

Because the traces lack utility bus locations, four-region coupling is a trace-driven benchmark: a predeclared feature-stratified round robin assigns class, GPU count, runtime, and submission time. Experiment 11 exhausts all 24 permutations, three penetrations, and two concentration controls. Experiment 18 uses fixed generator-bus electrical-role strata at native-load multiplier 0.90, so its cross-network result is conditional on the declared trace-to-bus scenario. Held-out energy ratios span 0.5845–2.6931. Equation (32) carries the fixed 6-MW/site component separately and calibrates the network scale on q_{99} ; both raw endpoints are therefore activation-eligible. The flexible-only 0.8901 capacity-safe factor is retained as a planning diagnostic, not an endpoint clip. PGLib and MATPOWER/PYPOWER provide topology, ratings, and costs [19], [34]; RTS-24 follows the published case [35].

B. Baselines, parameters, and inference

Experiment 2 compares eight meter/statistical baselines with the complete-ledger quantile, feasible projection, synthetic control, tail-risk counterfactual, the selected single projection, and risk-constrained verifier. payment-certified verifier is evaluated separately in Experiments 9 and 15 because its hull and N–1 objective differ. Feature equality is audited while execution truth is withheld. Deferral windows are 0, 4, and 512 slots; each declared region has a 118-MW flexible nameplate and a 6-MW fixed facility component. Six projection penalties and four risk reserves are validation-only, selected by nested contiguous folds before the locked split; four committed-response weights are fitted on earlier validation days. The gate commits 5 MWh, and payment is scored by an independent 40-segment evaluator. Moving three-day bootstrap replicates and all 2^{18} block-sign assignments use Holm correction [36], [37]. Sparse-system size grows from 5,760 variables/13,424 nonzeros at 96 slots to 36,480/85,488 at 608 slots (0.039–0.212 s).

The model-out check uses polar alternating-current optimal power flow (AC OPF) with active/reactive balance, voltage/reactive-generation bounds, and apparent-power ratings [19]. After the intact solve, every credible outage imposes, for each non-reference generator i ,

$$P_{Gi}^{(k)} = P_{Gi}^{(0)}, \quad i \notin \mathcal{G}_{\text{ref}}, \quad k \in \mathcal{C}. \quad (37)$$

The reference generator balances active losses, while $Q_G^{(k)}$ and voltages remain recourse. Apparent-power and voltage limits are checked after each fixed-plan solve, so (37) tests one shared active plan rather than reoptimizing active schedules. Non-reference bounds are fixed at the intact output $\pm 10^{-8}$ MW with a 10^{-6} -MW residual margin; the largest recorded deviation is 2.52×10^{-8} MW.

TABLE I
LOCKED 54-DAY MECHANISM-ISOLATION RESPONSE RESULTS (DAILY MEANS).

Method	nRMSE	False MWh	Under-credit	F_1
High-5-of-10	4.196	172.455	6.474	0.143
Ridge	2.761	109.817	5.306	0.233
Gradient Boosting	2.343	96.077	4.817	0.245
Extra Trees	3.350	133.961	6.804	0.171
Metadata Gradient Boosting	2.201	92.874	4.193	0.259
Ex-post Metadata Gradient Boosting	2.126	90.807	4.184	0.265
Ex-post Quantile Gradient Boosting	0.932	23.144	6.085	0.533
Synthetic Control	2.598	104.998	7.877	0.202
Feasible Quantile Projection	0.301	2.811	10.772	0.519
Single Feasible Projection	0.339	1.823	13.537	0.418
Tail-Risk Feasible Counterfactual	0.296	2.521	10.864	0.518
Risk-Constrained Convex Verifier	0.314	1.809	12.517	0.470

Experiments 1–2 test incentives and workload verification; 3–9 test signed settlement, transfer, risk, and N–1 value; 10–12 test AC/spatial transfer and continuous-cycle remuneration; 13–16 provide measured replay, the indexed witness, conversion intervals, and ledger reconciliation. Experiments 17–25 then freeze the gate information set, verify shared-plan AC feasibility, reconstruct the submitted-job witness, audit scale/coupling, replay an independent event, and enumerate all finite RTS-24 outages. Confirmatory comparisons use the same 54 locked days unless a deterministic network or ledger rule is stated.

V. RESULTS

A. Incentive boundary and independent verification

Experiment 1 solves the entire 8×7 DR-price/call-probability grid. The dual threshold (13) agrees with the independently solved strategic program in all 56 cells. At \$150/MWh and $q = 0.35$, strategic reference scheduling inflates the event baseline by 8.223 MWh. Of 55.452 MWh credited, 44.458 MWh is unsupported by physical reduction, giving a false-credit ratio of 0.802. The profitable region follows the derived marginal boundary and does not use a fitted classifier; the complete grid and its phase diagram are retained in the experiment output.

Experiment 2 evaluates twelve methods on the locked 54-day panel (Table I). The joint total-plus-CVaR fit gives nRMSE/ F_1 /false credit 0.314/0.470/1.809 MWh/day; CVaR-only gives 0.296/0.518/2.521, versus 0.339/0.418/1.823 for the single projection. The payment-contract cap is stored separately. Relative to feasible quantile, false credit falls 2.811 to 1.809 while nRMSE rises 0.301 to 0.314. Held-out changes are -0.025 , -0.014 , $+0.051$ for the joint fit and -0.043 , $+0.698$, $+0.100$ for CVaR-only. Removing both risk constraints gives 3.241 MWh/day; total-only, CVaR-only, and joint give 1.809, 2.521, and 1.809 MWh/day.

The independent trace-meter replay is a complementary deployment check: event-window MAE is 2.673/2.810/2.988 MW for feasible quantile/tail-risk counterfactual/risk-constrained verifier on the same 54 locked days (three-day block intervals in Experiment 20). It measures closed-DCGM alignment and remains separate from the mechanism-isolation response panel.

Closest domain baselines use identical arrivals, deadlines, capacities, event slots, and locked days. The incentive-compatible spatial-DR analogue gives nRMSE/false-credit/ F_1 3.068/114.776/0.137; flexibility and receding-horizon controls give nRMSE/false-credit/ F_1 0.339/1.823/0.418, while the Han-style joint-capacity, Chen-style coupled-regulation, and all-site

TABLE II
BASE-CASE AND COMPLETE N-1 SETTLEMENT ERRORS (USD/DAY).

Panel	Settlement	Trace MAE	Workload MAE	Trace median/overpay	Workload median/overpay
Base case	Nodal gross	360.580	421.677	—	—
Base case	Nodal exact net value	151.719	455.865	—	—
N-1	Base exact	139.221	167.367	79.79/30.42	121.01/107.69
N-1	N-1 linear	2.208	150.461	0.60/1.91	113.43/20.38
N-1	N-1 exact	0.990	150.955	0.87/0.01	113.43/20.18

exact comparators give 0.326/0.285 [16], [17]. These are structural analogues, with the receding-horizon translation from [13]; temporal-only/joint controls migrate 0/0.206 MWh. The ablation panel contains 810 globally solved schedules (binding shares 0.310/0.948; slope 1.003).

B. Nodal value and spatial response

Experiment 3 crosses thirteen counterfactuals with five settlement mechanisms. Table II isolates the mechanism with the trace-anchored reference and reports the locked workload rows. Exact signed nodal value has MAE \$151.719 versus \$360.580 for nodal gross response; the corresponding workload errors are \$455.865 and \$421.677. Signed netting reduces the trace error by \$208.190/day (Holm-adjusted $p = 3.89 \times 10^{-4}$), whereas uniform-to-nodal pricing changes it by $-\$1.463$ ($p = 0.0118$) and linear-to-exact value by $\$0.671$ ($p = 0.8026$). Workload effects are $-\$34.165$, $-\$1.191$, and $-\$0.023/\text{day}$, quantifying the mechanism-specific trade-offs on the same locked workload panel. All 5,616 interval/counterfactual instances satisfy (26); the largest violation is 2.03×10^{-11} dollars.

Experiment 4 applies the predeclared largest gross-minus-net offset (day 87), exposing the rebound component that positive-only local accounting omits.

C. Cross-network and N-1 security evidence

Experiment 5 evaluates 6,480 four-network outcomes; exact value improves the trace-anchored error in all 12 cells (mean \$1.913/day; eight Holm-adjusted tests below 0.05), with a \$0.050/day workload change. Experiment 6 solves 810 capacity/deadline schedules (binding shares 0.031–0.911; false credit 2.374–2.740 MWh).

Experiment 7 enumerates 16 four-site and 256 eight-site coalitions (maximum budget residual 5.68×10^{-14} dollars) and 1,296 states in the exact 20-participant panel. Experiment 8 enforces (31) for all 37 RTS-24 non-islanding outages: trace/workload MAE is \$139.221/\$2.208/\$0.990 and \$167.367/\$150.461/\$150.955 for base exact/N-1 linear/N-1 exact, with 1,000 p.u. maximum loading.

The complete replay scope is summarized in Table IV.

Experiment 9 tests (34) on 54 global programs with a validation-frozen seven-profile hull. Contiguous-fold baseline nRMSE selects the cap and independent validation N-1 MAE selects the payment target; locked outcomes are never opened. Held-out factors (0.584, 0.734, 0.996, 1.278, 2.693) have violations below 3×10^{-9} USD. In the order (payment-certified verifier, single, feasible-quantile, risk-constrained verifier), interior-factor MAE/overpayment is \$70.792/59.309/62.069/60.073 and \$2.947/7.727/6.954/7.502 USD/day. At q99, the corresponding MAE is \$214.314/179.816/188.315/182.297 and overpayment is \$9.289/24.307/21.883/23.615 USD/day. Experiment 15 retains

TABLE III
INFORMATION-BOUNDARY AND LOCKED RESPONSE CERTIFICATES (54-DAY MEANS). THE INDEPENDENT POLICIES USE A SUBMISSION-MASKED LEDGER, DISTINCT TARIFFS, AND NO VERIFIER OUTPUT; THE ROWS THEREFORE TEST CONTRACT/RESPONSE SEPARATION RATHER THAN A FIELD TREATMENT EFFECT.

Method	nRMSE	Gross false MWh	Payable MWh	Meter underpay	Recall	F1
Independent gate-committed response	2.606	145.506	227.116	94.055	0.436	0.358
Independent risk-constrained response	1.713	29.989	38.437	167.217	0.037	0.053
Complete-ledger risk-constrained verifier (mechanism isolation)	0.339	1.823	10.009	13.537	0.335	0.418

TABLE IV
COMPLETE DECLARED SECURITY-REPLAY COVERAGE.

Panel	Cells and certificate
Preventive DC N-1 (RTS-24)	$54 \times 8 \times 37$; 1,0000 p.u.
Shared-plan AC (IEEE-9)	3,888; 0.880/0.907/0.935 p.u.
Corrective/all-outage AC	864+5,400; 31,968 solves; all feasible

$q_{01} = 0.5845$ and $q_{99} = 2.6931$; the fixed/flexible conversion (32) activates both raw endpoints under the q99-calibrated scale without clipping, while 0.8901 is diagnostic. Unseen 0.80/1.20 MAE is \$16.186/\$24.285 (payment) and \$17.108/\$25.678 (feasible quantile), with neither factor certified; all stress panels use the same declared resource semantics.

Experiment 10 replaces the linear network with AC OPF [19]: all 864 connected base and 5,400 corrective IEEE-9/14 outage cells satisfy the limits. Experiment 18 gives zero deviation over 3,888 shared-active-plan cells (maximum loading 0.880/0.907/0.935 p.u. at 3/6/9%). Experiment 11 averages 26 regional assignments over three penetrations, four counterfactuals, and 54 days; RiskSafe/single/feasible-quantile MAE is \$105.250/\$211.555/\$315.490 and \$81.613/\$164.249/\$244.481 per day. Experiment 16 audits 66,769/28,413 training/test jobs (MAE 11.223 W, RMSE 15.818 W, $R^2 = 0.8994$); Experiment 12 reports event-window MAE \$26.937/\$15.969/\$5.325/\$5.325 and complete-cycle residual \$0.621/\$0.607/\$0/\$0 for feasible-quantile, payment-certified, RiskSafe, and single projection. The latter zero residuals are LP-precision accounting certificates, while event-window error remains the predictive test; all 54 contracts activate. All reported security cells are complete declared sets, not selected contingency subsets.

D. Independent trace and deployment-boundary audits

Immutable MIT scheduler/DCGM replay gives slot MAE 3.740 versus 5.678 MW at 100% coverage and a 71,128-job residual below 10^{-17} MWh; no utility event label is available, and trace-meter MAE is 2.988 MW (2.673 for feasible quantile). Experiment 23 uses a submission-masked ledger, predeclared tariff, and independent LP with no target, envelope, hull, or verifier output. Over 54 locked days, gate/independent responses score nRMSE 2.606/1.713, false credit 145.506/29.989 MWh/day, and $F_1 = 0.358/0.053$ (Table III); their nonzero trajectory differences certify information separation. Validation selects \$10/MWh, weight 1.0, and a 20-MWh/day budget; $\eta = 0.60$ is planning-only. Experiment 18 adds 9,648 fixed-active-plan AC-OPF outcomes at load multiplier 0.90 with zero voltage violation and maximum loading 1.000000 p.u.

E. Indexed workload counterfactual

Experiment 19 assigns one variable to each submitted job and admissible slot; the declaration window, 0.001-MW/GPU

TABLE V
EXACT INDEXED JOB-LEVEL COUNTERFACTUAL AND BINDING-CAPACITY CERTIFICATES.

Quantity	Value
Submitted / execution-matched jobs; job-slot variables	71,144 / 71,128; 12,296,675
Native / counterfactual event energy; gross/net; rebound (MWh)	0.194 / 0; 0.194/0.194 ; 0
Maximum job residual; minimum site slack (MWh)	0; 29.486
Binding stress cohort / capacity / service floor (jobs/MW/MWh)	689 / 0.001 / 0.004
Capacity-proportional / fixed-nameplate scale ($\times$)	2146.204 / 1.480
Job-to-network residual; coupled event slots (MWh / count)	0; 1,056
Secure N-1 value; RTS-24 outages / cells / solves	0.004293 ; 37 / 864 / 31,968
Maximum counterfactual post-contingency loading (p.u.)	1.0000

bound, regional entitlement, and 118-MW capacity yield 12,296,675 variables through an exact separable continuous-knapsack decomposition with typed residual checks. Native/counterfactual energy is 0.194/0 MWh, with zero residual and 29.486 MWh minimum slack (Table V). Experiment 22 verifies the invariant over all 1,056 event slots and finite RTS-24 outages, giving \$0.004293 per interval without aggregate reoptimization; Experiment 21 reports $2146.204 \times$ stress and $1.480 \times$ fixed-nameplate scales.

Experiment 25 independently validates the declaration boundary: 71,128/71,144 jobs match execution, requested-nameplate coverage is 1.000, and no committed window is infeasible. Its 689-job cohort binds 0.001 MW per region, meets a 0.004-MWh floor, has 43.75% binding slots, and slack above -10^{-19} MWh; this is separate from Experiment 19. Experiment 24 freezes two profiles and replays every 37 finite non-landing RTS-24 outages (864 cells; 31,968 solves); all succeed with maximum loading 1.000000 p.u. (Table V).

VI. CONCLUSION

This paper couples a provenance-bound workload verifier to signed nodal settlement. The validation-fitted total-plus-CVaR profile reaches $\text{nRMSE/false credit}/F_1 = 0.314/1.809/0.470$ MWh/day, while the slot-60 replay uses only committed declarations (Table III). The indexed witness has 12,296,675 variables, 0.194 MWh net reduction with zero rebound, and a zero-residual RTS-24 N-1 certificate at raw, fixed-nameplate, and capacity-proportional scales. A 689-job cohort binds 0.001 MW while meeting the 0.004-MWh floor; fixed demand remains outside conversion, so raw q_{99} is auditable and the flexible-only factor is diagnostic. Deployment requires signed co-located telemetry and topology access; causal utility validation remains external.

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